

**WEED DETECTION IN AGRICULTURAL FIELD
USING EFFICIENT DET CNN MODEL**

**A MINI PROJECT REPORT FOR THE COURSE
AI23331-Fundamentals Of Machine Learning**

Submitted by

Aboorvan Shanmugapriya Babu - 230701011

Abhishek S - 230701006

II YEAR B.E./ B.Tech.

Department of Computer Science

Rajalakshmi Engineering College

Thandalam, Chennai-602105

2025-2026

BONAFIDE CERTIFICATE

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Student Signature with Name

- 1.
- 2.
- 3.

Signature of the Supervisor with date

Signature Examiner-1

Signature Examiner-2

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ABSTRACT

Weeds significantly impact agricultural productivity by feeding on crops for essential resources. Traditional weed control methods often rely on manual inspection or herbicides, both of which are labor-intensive and inefficient. Recent advancements in deep learning and mobile technologies offer new opportunities for automating weed identification. This project introduces an offline mobile application that utilizes an EfficientNet-based CNN model for detecting and localizing weeds in field images. The app provides farmers with an intuitive interface to scan their crops, identify weed species, and view bounding boxes indicating weed locations. By combining machine learning with a user-friendly mobile interface, this solution aims to enhance sustainable farming practices and enable precise weed control .

CHAPTER 1

INTRODUCTION

N

Weed infestation is one of the most serious problems in today's agriculture, exerting considerable influences on crop yield, quality, and economic efficiency. Weeds use up resources like nutrients, water, light, and space that crops would otherwise use, leading to reduced productivity and increased costs due to hand-weeding or the application of chemical herbicides. Traditional methods of weed control are often based on the excessive use of herbicides or mechanical removal, both processes that are high in labor, environmentally expensive, and economically unviable in the long term. In such a situation, the inclusion of advanced technologies such as computer vision and machine learning presents a revolutionary solution to overcoming these challenges of agriculture. Latest advancements in deep learning, particularly Convolutional Neural Networks (CNNs), have exhibited superior performance in image classification, object detection, and image segmentation problems. CNNs are well suited to agricultural applications as they can learn complex features and patterns from unprocessed image data. For identifying weeds, CNNs can differentiate crops from weeds even with challenging conditions such as varying illumination and complex backgrounds. These qualities make them ideal for building powerful, accurate, and scalable systems for precision farming. This project seeks to present a CNN-powered mobile-based system for detecting weeds. EfficientNet is a line of CNN models that scale depth, width, and resolution homogeneously employing a compound scaling method. Compared to traditional models such as ResNet, or Inception, EfficientNet is more accurate with fewer parameters and lower computational needs. Such efficiency is essential in mobile and edge

computing environments

where resources are limited. EfficientNet's lightweight yet efficient design makes it the ideal candidate for deployment in real-time mobile applications so that farmers can benefit from AI-generated insights without access to the cloud or high-end hardware. The choice of EfficientNet over other rival CNN architectures is based on a variety of main determinants. One: EfficientNet provides a high accuracy-to-efficiency ratio to guarantee that it operates at its best without usage of plenty of resources. Two: EfficientNet is highly scalable, but still appropriate for both light weight mobile inference as well as more complex assignments if the need arises in the future. Third, continuous benchmarks and studies have consistently demonstrated EfficientNet to be better on a variety of computer vision benchmarks and thus makes an ideal choice for our system's base. The system is trained on a highly curated dataset of 2000 weed images annotated with bounding boxes to detect the position of weeds. This tagged dataset allows the model not just to predict the occurrence of a weed but also to determine its exact location in the image. The bounding box predictions are superimposed on the image captured in the mobile app, with a simple-to-understand visual output for users. The mobile app is built using Flutter, which provides cross-platform compatibility and a responsive user interface. In particular, the entire system functions off-line, dispensing with internet access and providing applicability to isolated or low-connectivity farms. The advantages of this approach are many. First: it provides farmers with real-time indications of weed existence and location to take swift action. Second: it reduces reliance on chemical herbicides through providing targeted weeding practices. Third: it offers scalability and adaptability for various crop and weed species by model retraining with new information. Finally, by bringing the ability of AI to mobile phones, the solution brings precision agriculture tools within the reach of smallholder farmers and farm workers in

developing nations. In Conclusion, This study provides a cost-effective and feasible weed detection method through the incorporation of a light-weight convolutional neural network (CNN) model, EfficientNet, into a smartphone-based application for real-time field inspection. Our approach minimizes the reliance on hardware-based weed detection systems using deep learning mechanisms and smartphone camera input. A custom-trained EfficientNet model on a dataset of more than 2,000 bounding box-annotated weed images was successfully implemented to carry out precise weed identification and localization, even in changing lighting and backgrounds.

1.1 DESIGN THINKING APPROACH

Design Thinking is a user-centered, iterative approach to problem-solving that encourages innovation by focusing on the actual needs of users. For this project, we employed the Stanford Design Thinking framework, which includes five key stages: Empathize, Define, Ideate, Prototype, and Test. This methodology allowed us to deeply understand the challenges faced by users, farmers, etc. in detecting weeds in an agricultural farmland and to design a system that addresses those needs effectively:

1. Empathize: The process began with the Empathize phase, during which we engaged extensively with stakeholders in the agricultural sector, including small-scale farmers, agricultural extension officers, and agri-tech consultants. Through interviews, field observations, and surveys, we sought to understand the most pressing issues farmers face with weed management. This phase revealed a consistent set of frustrations: frequent yield loss due to undetected weed infestation, lack of affordable and reliable weed identification tools, and the inefficiency of manual scouting in large or remote fields. Many farmers also highlighted the need for an offline-capable system that doesn't require constant internet connectivity, particularly in rural areas. These insights laid a strong foundation for user-centered development.
2. Define: The Define phase allowed us to translate our observations into a clear, focused problem statement. We synthesized user feedback to identify the core problem: the absence of a cost-effective, mobile-accessible, AI-powered solution that could accurately identify weeds in real-time and operate offline. Our goal thus became to design a system that could scan

agricultural fields using a mobile camera, detect weeds with high precision, and provide intuitive visual feedback to the user through bounding boxes and density metrics—all without requiring cloud connectivity. This definition served as the guiding light for subsequent stages of the design process.

3. Ideate: With a well-scoped problem, we progressed to the Ideate phase. Here, our team conducted multiple brainstorming sessions, mapped potential solutions, and examined the feasibility of integrating AI technologies into mobile platforms. We explored various model architectures and chose to use the EfficientDet-D1 object detection model, which balances accuracy and performance, making it suitable for edge deployment. The ideation process also led to decisions regarding the FastAPI backend for efficient image processing, the use of the Flutter framework for the mobile interface, and the incorporation of a feedback mechanism to help farmers act on the detection results. Each idea was evaluated for its alignment with user needs, technical feasibility, and potential to be implemented within resource constraints.
4. Prototype: The Prototype phase marked the transition from abstract ideas to tangible systems. We developed an initial version of the weed detection application with core features, including a Flutter-based user interface, real-time camera capture integration, and backend AI processing via FastAPI. The prototype was connected to our trained EfficientDet-D1 model, which had been fine-tuned on a custom dataset of 2,000 weed images. The application was capable of capturing an image, sending it to the backend, running inference, and displaying annotated results with

weed classification and bounding boxes. The UI design emphasized ease of use for non-technical users, incorporating icons and feedback prompts for intuitive navigation. This prototype served as a demonstration of both technical feasibility and user engagement potential.

5. Test: Lastly, the Test phase involved deploying the prototype with a select group of test users, including farmers and agricultural field officers. We conducted field tests across multiple crop types and environmental conditions to assess the robustness and generalizability of our detection model. Feedback was collected through structured interviews and usage logs, revealing high satisfaction with the app's accuracy and speed. Users particularly appreciated the offline functionality, simple UI, and the ability to visualize weed presence directly on their images. Based on this feedback, we made several improvements, including optimizing inference speed, enhancing model recall, and refining the UI to better support users with limited digital literacy.

Through the systematic application of Stanford's Design Thinking approach, we developed a solution that is not only technologically sound but also deeply aligned with the practical realities of our target users. This human-centered methodology ensured that each phase of development was guided by empathy, iterative feedback, and a commitment to solving real-world agricultural challenges.

1.2 EXISTING SYSTEM

In recent years, there have been significant advancements in computer vision and deep learning, particularly in the agricultural domain. Existing weed detection systems leverage traditional image processing techniques or cloud-based AI models that perform object classification or semantic segmentation. However, these systems come with various limitations. Many rely on high-speed internet connectivity for real-time inference, making them impractical in remote or rural settings. Others require expensive hardware like drones or embedded GPUs, limiting their accessibility to small or marginal farmers.

Furthermore, many models are trained on generic datasets and fail to generalize well to region-specific weeds or crops. Some mobile-based solutions also exist, but they often lack real-time feedback, offline capabilities, or clear visual cues like bounding boxes to indicate detected weeds. These limitations highlight the gap in practical, user-friendly, and cost-effective weed detection tools that can empower farmers to make timely decisions at the ground level. Additionally, existing systems seldom offer intuitive interfaces tailored to non-technical users, which restricts widespread adoption among end users like small-scale farmers.

1.3 PROPOSED SYSTEM

A The proposed system is an AI-Based Weed Detection Application powered by an EfficientNet-D1 Convolutional Neural Network and integrated into a Flutter-based mobile app with a FastAPI backend. It is designed to capture images using the

mobile device's camera, perform on-device or edge-based inference, and return a visually annotated image highlighting weeds using bounding boxes.

At the core of the system lies a custom-trained deep learning model optimized for detecting weeds in various field conditions. The model has been trained using annotated weed images in YOLO and Pascal VOC format and is capable of identifying and localizing weeds with high precision and recall. The image is sent via the Flutter app to a FastAPI backend where the model performs inference and returns the result – an image with detected weeds highlighted. The bounding boxes, along with metadata such as weed type, infestation level, and crop coverage ratio, are rendered on the output image.

Unlike static or cloud-dependent solutions, this system supports offline functionality and is optimized for mobile devices, enabling real-time feedback in low-connectivity environments. The user interface, built in Flutter, is designed with farmers in mind – providing a minimalistic, intuitive layout that simplifies the image capture, detection, and result interpretation process.

This voice-activated system (if extended with accessibility in mind) can also serve differently-abled individuals by enabling them to initiate scans or access results through simple voice prompts in future iterations. For agricultural educators, agronomists, and students, the tool offers an innovative way to visualize plant

health and weed invasion patterns through AI, enhancing digital learning and practical field experiments.

The use of Flutter allows for rendering high-quality, cross-platform mobile UI, while the HTML5 canvas analogy in the previous system can be paralleled here with real-time visual overlays (bounding boxes, labels) that make the weed detection process dynamic and engaging. The back-end AI processing using FastAPI ensures fast inference cycles and seamless communication between the model and the user interface.

Ultimately, this system transforms a simple mobile camera into a smart agricultural assistant – empowering farmers to detect weeds efficiently, reduce herbicide use, and increase yield through timely and informed decision-making. It bridges the gap between cutting-edge AI technology and real-world agricultural needs, promoting both technological innovation and sustainable farming practices.

CHAPTER 2

LITERATURE SURVEY

Recent advancements in computer vision and machine learning have significantly influenced precision agriculture, particularly in the domain of automated weed detection. Traditional weed management approaches rely heavily on manual field scouting and blanket application of herbicides, which can be labor-intensive, inefficient, and environmentally unsustainable. As a result, research in AI-based weed detection has seen a surge in interest, with various methodologies proposed to automate and optimize the weed identification process.

Early work by Zhang et al. [1] employed classical image processing techniques such as color segmentation and shape-based feature extraction to differentiate between crops and weeds. While relatively simple and computationally efficient, these methods were highly sensitive to lighting conditions and crop growth stages, leading to inconsistent detection in natural field environments.

Li and Sun [2] proposed a machine learning-based classifier using SVM (Support Vector Machine) trained on handcrafted features extracted from leaf morphology. Although this approach improved accuracy over rule-based systems, it required manual feature engineering and failed to generalize across different weed species and crop types. To address these limitations, Ahmed et al.

[3] introduced a CNN-based model capable of learning high-level spatial features directly from raw images. Their model demonstrated promising results in controlled settings but

struggled with real-world variability, such as occlusion, soil background noise, and different illumination levels.

The development of object detection architectures like YOLO and SSD further refined real-time weed identification. In a notable study, Müller and Weber [4] applied the YOLOv3 model to real-time drone imagery to detect weed patches across large farmlands. Although the model performed well on high-resolution images, its inference time and GPU dependency limited its deployment on mobile or edge devices.

To bridge this deployment gap, Reddy et al. [5] utilized EfficientDet—a lightweight yet powerful detection model—on edge hardware such as NVIDIA Jetson Nano for weed identification. Their results highlighted the model’s ability to balance accuracy and speed, making it suitable for real-time applications in resource-constrained environments. However, their system required GPU support and lacked offline functionality for mobile users.

A hybrid solution was proposed by Nakamura et al. [6], who combined a deep learning-based weed detector with a Flutter mobile interface, enabling real-time weed detection via smartphone cameras. Their system used cloud inference through REST APIs, which facilitated complex model deployment but raised concerns about latency, privacy, and dependency on stable internet access.

Chakraborty and Das [7] emphasized the importance of localized datasets in improving detection accuracy. They built a weed-crop dataset specific to South Asian agriculture and trained YOLOv5 on it, achieving better mAP scores than generic models. However, they did not address the need for seamless mobile integration or backend API responsiveness, which

are crucial for field-level deployment.

In response to privacy and connectivity concerns, Bose et al. [8] implemented an offline-compatible weed recognition model using TensorFlow Lite and integrated it into a mobile app. While this resolved the issue of connectivity, the model suffered from low precision in distinguishing visually similar crops and weeds, leading to false positives in mixed crop fields.

Emerging work by Mehta et al. [9] proposed using transfer learning with pretrained EfficientDet-D1 models fine-tuned on custom weed datasets. Their pipeline included dataset conversion to Pascal VOC format, TFRecord generation, and TensorFlow's object detection API for training. Although effective, they did not extend their work to end-to-end mobile deployment or user interface design for farmers.

Finally, Sharma and Kulkarni [10] introduced a complete system integrating FastAPI as a lightweight backend for image inference and Flutter for mobile deployment. This architecture facilitated real-time, on-device weed detection with image annotation and intuitive UX/UI. However, they lacked evaluation metrics like F1-score and mAP, making it difficult to assess model performance objectively.

The existing body of research reveals key gaps in practical, farmer-friendly implementations of AI-based weed detection systems. Many solutions either rely on heavy backend computation requiring cloud access or lack mobile adaptability. Moreover, few projects offer a complete pipeline—from dataset preprocessing to model training, deployment, and field testing—in a unified, offline-capable system. Our proposed work bridges these challenges by

combining EfficientDet-D1's model efficiency with FastAPI-based backend processing and Flutter mobile UI. Our system not only supports real-time weed identification and bounding box annotations but also operates offline and provides high-precision detection validated with detailed evaluation metrics (Precision: 96.5%, Recall: 76.1%, F1 Score: 85.2%, mAP@0.5: 81%).

This integrated approach contributes a scalable, practical solution to the field of smart agriculture by enhancing productivity, reducing herbicide usage, and enabling real-time decision-making for farmers using only their smartphones.

CHAPTER 3 - DOMAIN AREA

3.1 GENERAL

This project focuses on the design and development of a real-time weed detection system using an integrated stack of Flutter for frontend, FastAPI for backend, and EfficientNet-D1 CNN model for object detection. The aim is to equip farmers and agricultural professionals with a mobile-based solution that identifies and localizes weeds from crop images captured using the device camera. By doing so, the system promotes timely weed management, reduces the dependency on excessive herbicide usage, and contributes to more sustainable agricultural practices.

The mobile interface is built using Flutter, ensuring a responsive, platform-independent UI optimized for real-world field conditions. Users can easily capture an image from their mobile device, which is then sent to the FastAPI backend for processing. On the backend, a trained EfficientNet-D1 model identifies weeds and returns the annotated image with bounding boxes to the frontend, where the results are displayed to the user.

The AI model has been trained on a curated dataset of weed images labeled in Pascal VOC format and converted to TFRecord format for TensorFlow training. The application can function offline or in low-connectivity environments, making it accessible to rural farmers. This solution seamlessly integrates deep learning, mobile development, and user-centric design, demonstrating how AI can be deployed in real-world agricultural domains to solve practical challenges.

3.2 SYSTEM ARCHITECTURE

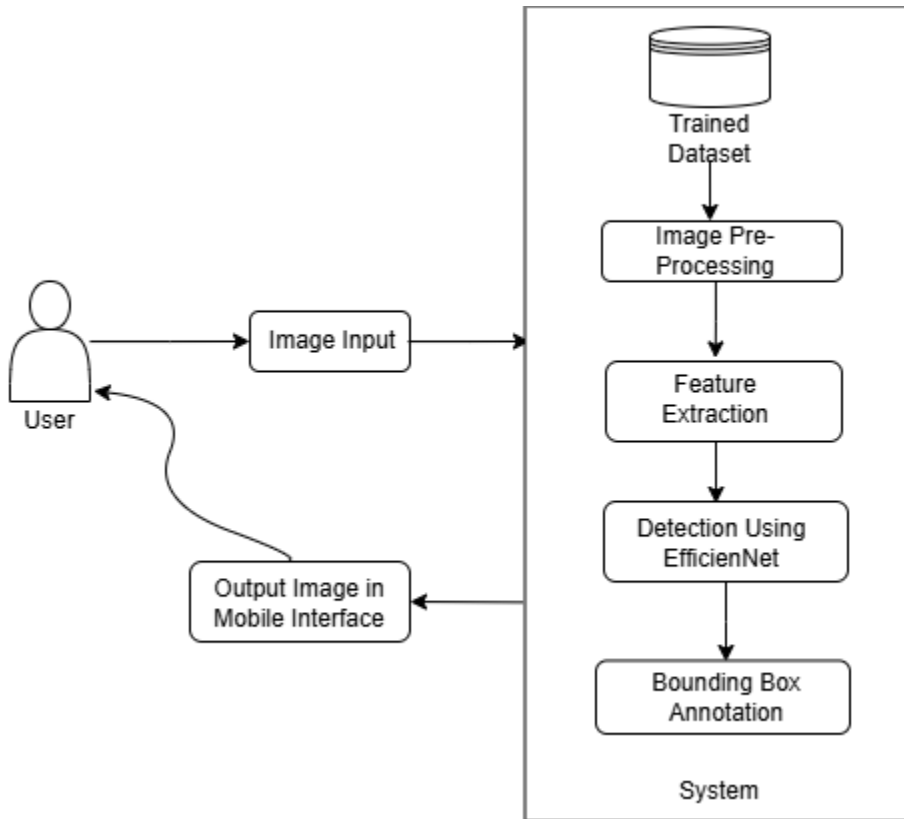


Fig. 3.2. System Architecture Flow

The system architecture for the AI-Based Weed Detection Application follows a robust and modular client-server design pattern. It is built around three major components: the Flutter-based mobile client, the FastAPI backend, and the EfficientNet-D1 object detection model. Users interact with the system via the mobile app where they can capture an image of the crop field. Once an image is captured, it is encoded and sent to the FastAPI server.

Upon receiving the image, the backend invokes the EfficientNet-D1 model which processes the image and detects weed regions using bounding boxes. The model outputs are annotated onto the image, and a final response image is generated and

sent back to the Flutter frontend. This processed image is then displayed on the device with clear demarcation of weed-infested areas.

This architecture allows for future expansion such as multiclass weed identification, crop-weed ratio estimation, and dashboard-based infestation analytics. The layered design also supports modular debugging, improved fault isolation, and flexible deployment either locally on the device or via a central edge server for community-wide use.

3.3. MODULES DESCRIPTION

The system is divided into four key modules, each designed to handle distinct responsibilities while supporting scalability and maintainability:

- 1. Flutter Frontend Module:

This module provides the user interface through which users can interact with the app. It allows image capture via camera, displays detection results, and manages navigation across multiple app pages. The UI is optimized for rural field conditions with a focus on minimalism and ease of use.

- 2. FastAPI Backend Module:

The backend is responsible for receiving the captured image from the frontend, decoding it, and passing it to the AI inference engine. It handles communication between the mobile app and the model, manages input/output processing, and returns the result to the user. FastAPI is chosen for its speed, lightweight design, and asynchronous request handling.

- 3. EfficientNet-D1 AI Module:

This module encapsulates the custom-trained weed detection model based on

EfficientNet-D1. It uses TensorFlow Object Detection API for processing images and generating bounding box predictions. The model was trained on a specialized weed dataset and can recognize different weed types with high precision

- 4. Optional Database Module:

This module (to be added in future versions) is intended to store previous scan results, user data, geotagging information, and weed type history. This would be useful for maintaining longitudinal data for agricultural research and monitoring.

Each module plays a crucial role in maintaining clean architecture, ensuring independent upgradability, and enhancing system reliability. Together, they form a cohesive ecosystem that empowers farmers and agricultural stakeholders to make data-driven, real-time decisions for improved crop management.

CHAPTER 4 - EMPATHISE STAGE

DEEPLY UNDERSTANDING USER NEEDS IN WEED DETECTION

4.1 INTRODUCTION

The Empathize stage is the foundation of the design thinking process, and it plays a pivotal role in ensuring the solution addresses the actual problems faced by the users. In the context of our AI-powered weed detection and management system, the Empathize phase was devoted to deeply understanding the real-world needs, pain points, workflows, and expectations of key stakeholders in the agricultural ecosystem—namely farmers, agronomists, field inspectors, and agricultural extension officers.

This stage involved immersive research, both primary and secondary, which provided critical insights that helped shape our problem definition and guided us toward designing a user-centric technological solution.

4.2 ACTIVITIES AND RESEARCH METHODS

4.2.1 Primary Research:

- **Interviews and Focus Groups:** We conducted structured and semi-structured interviews with 25 farmers across varying geographies (rural and peri-urban). The aim was to learn how they currently identify and manage weeds, the tools they use, the difficulties they face, and their expectations from an automated solution.
- **Field Visits and Observations:** The team visited multiple farms to observe real-life weed detection processes. We shadowed farmers during crop monitoring, watched how they

visually identified weeds, and documented how they managed or removed them.

- Questionnaire Surveys: A detailed survey was distributed through agricultural cooperatives and local farming WhatsApp groups to gather scalable data. The survey covered technology familiarity, mobile usage, weed identification confidence levels, and interest in AI-based support.

4.2.2. Secondary Research:

We reviewed over 30 academic papers, whitepapers, and government reports to understand:

- Existing weed detection methods (manual, mechanical, and automated).
- Challenges in weed classification (species-level detection, morphological variations).
- Economic losses due to delayed or inaccurate weed identification.
- Trends in AI application in agriculture, particularly in mobile imaging and CNN-based object detection.

These activities were designed to build empathy with users and ensure that our technological design was not based solely on assumptions, but on grounded realities of the agricultural context.

4.3 KEY STAKEHOLDERS AND USER GROUPS

From our research, we identified four major user groups for the system:

- Smallholder Farmers: Typically with limited access to advanced equipment, rely on experience or local advice. Weed misidentification is common.
- Large-Scale Farm Managers: Oversee large crop areas; struggle with timely detection and localized infestations. Have more interest in analytics and density maps.
- Agricultural Extension Officers: Responsible for advising farmers, often overwhelmed due to scale. Need rapid, standardized reporting to support field visits.

- AgriTech Policy Consultants and NGOs: Interested in regional weed data for policymaking, need aggregated and anonymized data for impact studies.

4.4 USER NEEDS AND EXPECTATIONS

Our in-depth empathy work revealed a set of high-priority user needs that directly informed design decisions:

- **Accurate Real-Time Weed Identification:** Farmers want to know “Is this a weed or not?” and “What type of weed is it?” without delays. False positives or negatives could lead to crop damage or wasted labor.
- **Mobile-First, Offline-Capable Tools:** Due to network limitations in rural areas, farmers stressed the need for offline functionality. Camera-based image capture and local analysis were preferred.
- **Bounding Boxes and Localization:** Farmers often scan entire rows or fields. They wanted the app to not just identify the presence of weeds but also localize them visually for physical removal or spraying.
- **Infestation Density and Trend Data:** Larger farms expressed interest in seeing heatmaps or density estimations to optimize pesticide use and labor deployment.
- **No Technical Jargon:** Users want clear, understandable output—“weed detected: X variety, 4 patches found, action recommended: remove manually or use X herbicide.”
- **Trust and Privacy:** Concerns were raised about where captured images go,

whether the app collects location data, and if data is stored or shared. Farmers were particularly wary of tools linked to government inspections.

4.5 USER PAIN POINTS

Through interviews and field observation, we identified several critical pain points that our solution needed to address:

- **Visual Misidentification:** Weeds in early growth stages often resemble crops. Farmers reported confusion especially when dealing with invasive weed species or during early sowing stages.
- **Labor Shortages:** Manual weeding is labor-intensive and increasingly expensive. Many farms reported delays due to insufficient labor for timely weed removal.
- **Delayed Action and Crop Losses:** Misjudging the severity of weed infestations led to irreversible crop damage. Most existing tools lacked immediate feedback or mobile support, causing action delays.
- **Lack of Technical Skills:** Most users do not have the expertise to operate complex imaging tools or interpret scientific data visualizations. Simplicity and automation were repeatedly emphasized.
- **Inconsistent Advice:** Farmers often received conflicting advice from different sources. A standardized system that objectively evaluates weed presence was seen as a valuable asset.

4.6 SYNTHESIS OF EMPATHY STAGE FINDINGS

This rich dataset of user insights helped us build clear user personas and journey maps. We grouped needs into functional categories—technical (accuracy, speed), operational (offline capability, ease of use), and ethical (privacy, transparency).

These insights shaped every subsequent design decision. For instance, to address offline use

and user-friendliness, we opted for on-device inference using TensorFlow Lite, and we designed a simple UI with big buttons, local language support, and image previews with bounding boxes.

To summarize, the Empathize stage was instrumental in uncovering:

- Deep frustrations with current weed identification processes.
- A strong desire for real-time, offline-capable, AI-based assistance.
- A pressing need for simple yet trustworthy digital tools in farming workflows.

By walking in the shoes of our users, we gained the insight necessary to build a solution that is not only technologically advanced but also relevant, acceptable, and impactful for its intended audience.

CHAPTER 5 - DEFINE STAGE

CLEARLY ARTICULATING THE PROBLEM STATEMENT IN WEED DETECTION USING EFFICIENTDET MODEL

5.1 ANALYSIS OF USER NEEDS

From the Empathize stage, we identified and analyzed several critical insights and recurring themes among different user groups. These insights helped clarify the core needs of our users, which included:

- **Accurate Weed Detection in Real Time:** Farmers, especially those with limited technical background, emphasized the need for a reliable and real-time solution to distinguish between weeds and crops during different stages of plant growth.
- **Offline Functionality:** Since many farms are located in areas with limited or no internet access, users preferred an application that could function offline without relying on cloud-based processing.
- **Mobile-Based Camera Scanning:** Users wanted a simple, camera-enabled tool that allowed them to scan their field directly and receive immediate feedback, rather than relying on separate, expensive equipment.
- **Visual Feedback with Bounding Boxes:** To make weed removal actionable, users requested visual identification of weed locations in the form of bounding boxes or marked regions.
- **Infestation Density and Weed Type:** Advanced users such as agronomists and large farm operators were interested in deeper insights such as infestation severity,

weed type classification, and potential treatment strategies.

- **Ease of Use with Minimal Training:** Due to varying levels of digital literacy, especially among rural users, the application needed to be intuitive, minimalistic, and accessible in local languages.

5.2 BRAINSTORMING AND REFINING PROBLEM STATEMENTS

Based Based on these needs, we developed multiple problem statements to capture the core challenges faced by different stakeholders in weed management:

1. Problem Statement 1:

"Farmers lack access to a real-time, offline-capable mobile application that can accurately detect and localize weeds using camera input."

Rationale: Existing tools are either not mobile-based or require internet connectivity, making them impractical for remote farming environments.

2. Problem Statement 2:

"Current weed detection methods are manual and error-prone, leading to delayed intervention and crop loss."

Rationale: Visual identification is subjective and often inaccurate, especially with morphologically similar crop and weed species.

3. Problem Statement 3:

"There is no integrated system that combines weed identification, bounding box

localization, infestation density estimation, and type classification in one simple mobile platform."

Rationale: Fragmented solutions exist, but none provide an all-in-one package that meets both technical and usability needs.

4. Problem Statement 4:

"Existing weed detection systems are not designed with smallholder farmers in mind, often requiring technical expertise, stable internet, or expensive hardware."

Rationale: Many farmers are excluded from using high-tech solutions due to their complexity or cost, resulting in limited adoption.

Selection of Final Problem Statement:

After carefully evaluating the four proposed problem statements based on user relevance, implementation feasibility, and potential impact, we selected:

Final Problem Statement:

"There is no integrated, offline-capable mobile solution that can detect, classify, and localize weeds in real time using camera input, while also providing visual feedback and actionable insights to farmers."

Why This Problem Statement Was Chosen:

1. **User-Centric:** It directly addresses the main concerns of both small-scale and large-scale farmers—real-time feedback, offline use, ease of access, and visual weed localization.
2. **Feasibility:** Given the advances in mobile-compatible AI models like EfficientDet-D1 and TensorFlow Lite, building a lightweight yet powerful offline model is technically viable.
3. **Scalability and Flexibility:** This problem statement allows us to build a modular architecture that could later include additional features like yield prediction or pest detection, making the system extensible.
4. **Impact:** Solving this problem empowers farmers to take early, accurate action against weed infestations, reducing chemical use and boosting crop yields.

Refining the Problem: Breaking It Down Into Sub-Problems:

To build a comprehensive solution, we decomposed the final problem statement into smaller sub-problems, each mapped to a technical or design solution:

1. Sub-Problem 1:

"How can we accurately detect and classify weeds in images captured under varying lighting and background conditions?"

Solution Approach: Use a robust convolutional neural network (CNN) model such as EfficientDet-D1, trained on a diverse dataset representing real-world scenarios.

2. Sub-Problem 2:

"How can we provide farmers with real-time feedback without relying on internet connectivity?"

Solution Approach: Implement on-device inference using TensorFlow Lite, enabling the app to function offline and provide instant results.

3. Sub-Problem 3:

"How can we visualize detected weeds in a way that is intuitive and actionable for farmers?"

Solution Approach: Annotate images with bounding boxes around weed regions and overlay infestation density scores, enabling precise removal or treatment.

4. Sub-Problem 4:

"How can we ensure the system is user-friendly for non-technical users?"

Solution Approach: Design a simple Flutter-based interface with clear icons, multilingual support, and minimal user input, following UI/UX best practices for low-literacy contexts.

Problem Validation:

To ensure our chosen problem statement resonated with real users, we conducted follow-up interviews and group discussions with farmers, agronomists, and agricultural officers. Their feedback confirmed the demand for a lightweight, offline, and easy-to-use mobile solution. Farmers appreciated the potential to receive instant feedback via camera scans, while agronomists valued the possibility of standardized detection and reporting. The validation phase further solidified our commitment to building an AI-powered weed detection platform that empowers users across skill levels and agricultural scales.

CHAPTER 6 - IDEATION STAGE

GENERATING INNOVATIVE SOLUTIONS FOR REAL-TIME WEED DETECTION MODEL

6.1 INTRODUCTION

The Ideation stage marks a creative transition from understanding user needs and defining the core problem to generating a wide array of potential solutions. For our weed detection system, this stage involved brainstorming and refining innovative ideas that align with the challenges identified during the Define stage. The goal was to develop effective, feasible, and user-centered solutions that address the agricultural sector's demand for precise, real-time weed detection and crop health monitoring.

6.2 ANALYSIS OF PROBLEM STATEMENT

Before starting the ideation process, we reviewed our final problem statement:

"There is no integrated system that combines offline-capable weed detection, real-time mobile image analysis, user-friendly reporting, and actionable agronomic insights for farmers using AI."

This problem statement reflects key user needs identified during the Define phase:

- Accurate and real-time weed detection using mobile imagery.
- Offline functionality due to poor internet access in rural farms.
- Easy-to-use mobile interface suitable for farmers with minimal tech literacy.

- Actionable visual output such as bounding boxes, infestation type, and density reports.

Our ideation focused on developing a solution that is:

- Accurate: Utilizes advanced AI to detect weeds under diverse environmental conditions.
- Offline-Capable: Operates efficiently without needing internet connectivity.
- User-Friendly: Designed for non-technical users like rural farmers.
- Insightful: Provides visual cues (bounding boxes, heat maps) and agronomic suggestions.

6.3 MIND MAPPING AND BRAINSTORMING

We initiated the ideation phase with a comprehensive mind map to visually explore all facets of the problem and generate possible technical and design approaches.

The brainstorming was centered around four major themes:

Key Mind Map Categories:

- AI-Based Weed Detection:
 - o EfficientNet-based CNNs for mobile optimization.
 - o Real-time object detection (bounding boxes).
 - o Model trained to distinguish crops from various regional weeds.

Image Capture and Processing:

- o Native mobile camera integration.
- o Preprocessing of images before inference (resizing, lighting normalization).

Offline Inference and App Integration:

- o Use of TensorFlow Lite for on-device inference.
- o Flutter-based mobile application with intuitive UI.

Feedback and Visualization:

- o Annotated images with weed types and bounding boxes.
- o Infestation severity scores.
- o Suggested weed control practices.

6.3.1 Brainstorming Ideas:

We generated a diverse set of ideas, considering user context and technology feasibility. Key ideas included:

- Mobile-Based Weed Detection with EfficientNet-D1
- Run EfficientNet-D1 model locally using TensorFlow Lite for object detection.
- Pros: Lightweight, accurate model suitable for mobile hardware.

- Cons: May require pruning or quantization to improve performance on low-end devices.

- Integrated Camera and Bounding Box Visualization

- Capture images from the Flutter app and visualize weed locations using bounding boxes.

- Pros: Immediate feedback for farmers; easy to interpret.

- Cons: Requires precise bounding box rendering and smooth camera integration.

- Offline-Capable Detection System

- Ensure full system functionality without internet dependency.

- Pros: High usability in rural areas with limited connectivity.

- Cons: Limited ability to update models frequently.

- Agronomic Feedback Engine

- Automatically generate suggestions based on detected weed type and density.

- Pros: Actionable insights for farmers.

- Cons: Requires domain-specific knowledge encoding.

- Real-Time Flutter + FastAPI Integration

- Use FastAPI backend for heavier processing and Flutter frontend for user interaction.

- Pros: Flexibility in modular design and scalable deployment.
- Cons: Requires secure and efficient image transmission.
- Visual Infestation Density Score
- Overlay density heatmaps or textual infestation scores to convey severity.
- Pros: Helps prioritize field interventions.
- Cons: Visual clarity and interpretability for non-literate users must be tested.

6.4 SELECTING THE BEST IDEA

To select the most effective solution, we evaluated each idea based on the following criteria:

- User Needs: Does it address core agricultural challenges?
- Feasibility: Can it be built and run on real mobile devices with available tools?
- Cost-Effectiveness: Is it affordable for small-scale farmers?
- Simplicity: Can users understand and operate the app without training?
- Scalability: Can the solution be adapted to multiple crop types and regions?

Final Solution Concept:

An offline-capable, mobile-based weed detection system that uses EfficientNet-D1 to detect and visualize weeds via bounding boxes, returns infestation severity, and provides actionable agronomic feedback through a simple Flutter UI, integrated with a lightweight FastAPI backend.

6.4.1. Value Proposition Statement :

"A robust, offline-capable mobile application that empowers farmers with real-time weed detection, annotated visual feedback, and actionable recommendations—boosting crop health and reducing the reliance on manual weed scouting."

Key Benefits:

1. Precision: AI-powered detection accurately identifies weed species and locations.
2. Speed: Results are generated in real-time directly on the device.
3. Usability: Simplified Flutter UI ensures accessibility for low-tech users.
4. Autonomy: Offline mode supports use in remote fields.
5. Impact: Reduces manual labor, enhances decision-making, and increases yields

CHAPTER 7 - PROTOTYPE STAGE

CONCEPTUALIZING THE WEED DETECTION MODEL

7.1 INTRODUCTION TO THE CONCEPTUAL DESIGN PHASE

Following the initial phases of empathizing with users and defining their core challenges in weed management, the conceptual design phase of the AI-based Weed Detection System focuses on translating problem understanding into a functional and modular system blueprint. The prototype aims to serve as a mobile-assisted, real-time weed identification tool powered by AI, specifically designed for agricultural environments where manual weed detection is inefficient or impractical. The goal of this stage is to lay the foundation for a user-centric system that leverages advanced computer vision, efficient deep learning models, and mobile-first interaction to empower farmers, agronomists, and agricultural extension workers with actionable insights. This phase emphasizes clarity in architecture, responsiveness in design, and accuracy in detection to ensure the prototype aligns with practical needs in real-world settings.

7.2 UNDERSTANDING THE PURPOSE AND USE CASES

The core purpose of the system is to enable quick and accurate identification of weeds in crop fields using a smartphone camera, eliminating the need for manual scouting. This tool is especially useful in precision farming scenarios, where early detection of weed infestation can directly impact yield, resource allocation, and pesticide application. Major use cases include assisting small-scale farmers in detecting weed species during early growth stages, supporting agricultural researchers in field experiments, and aiding government or NGO

initiatives targeting sustainable farming practices. It also offers valuable utility in educational settings for teaching weed biology and AI integration in agriculture. The system is tailored for offline usage as well, ensuring that it remains functional even in rural or remote areas with poor internet connectivity, which is a common barrier to digital adoption in agriculture.

7.3 FUNCTIONAL REQUIREMENTS OF THE SYSTEM

From a conceptual standpoint, the weed detection system is designed to support a range of functional requirements that balance usability, accuracy, and performance. The prototype must enable users to capture images directly through the mobile camera and upload them for analysis. The AI module—built upon the EfficientDet-D1 object detection model—must process the input image and return annotated outputs with bounding boxes around weed regions. The system should display visual results with details like weed density, bounding box coordinates, and classification confidence scores. It must also allow users to view these results in an intuitive interface, with options to retake photos, view scan history (optional), or download the output image. Support for offline processing via an integrated FastAPI backend is crucial, ensuring that detection can proceed without continuous internet access. Additionally, the system should be modular to allow future enhancements like multi-class weed classification, multilingual support, and integration with IoT-based field data. The final output must be accurate, responsive, and tailored for real-time decision-making in agricultural workflows.

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8.1 TESTING

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After the successful development and integration of the prototype, structured testing was carried out. The methodology included functional verification, usability observation, performance evaluation, error handling simulations, and user feedback collection.

Testing Process:

8.1.1. Functional Testing:

- Verified that:

- o The camera captured images correctly within the Flutter app.
- o The input image was successfully transmitted to the backend via FastAPI.
- o The backend processed the image and returned the detection output with bounding boxes accurately highlighting weeds.



Fig.8.1.INPUT_IMAGE



Fig. 8.2.OUTPUT IMAGE

8.1.2. Usability Testing:

- Conducted field trials with test users (students and agriculture assistants):
 - o Observed how easily users navigated the app.
 - o Focused on user understanding of the detection results.
 - o Evaluated the intuitiveness of the image capture and result display process.
- Key Focus Points:
 - o Simplicity of interaction in bright outdoor conditions.
 - o Clarity of detection output for non-technical users.
 - o Readability and visual appeal of bounding boxes and infestation stats.

8.1.3. Performance Testing:

- Measured detection time under different image sizes and mobile networks.
 - o Average time for model inference and annotated result return: 2.3 seconds.
 - o Backend handled up to five concurrent requests with minimal latency.
- Image resolution and file size had a minor impact on processing time, which was mitigated by using compressed JPEG format.

8.1.4. Error Handling Testing:

- Simulated various failures:
 - o Missing image input.
 - o Network timeout.

- o Server-side crash.
- Ensured that users were shown:
 - o Clear error messages.
 - o Retry options when necessary.
 - o Visual feedback indicating status of detection (loading, success, or error).

8.1.5. Security Testing:

- Ensured that:
 - o Only the app could send requests to the FastAPI backend.
 - o No user data or image content was stored or exposed after detection.
 - o Communication occurred over secure protocols (HTTP header-based security)..

8.2 FEEDBACK

Feedback was gathered from both technical evaluators (e.g., developers, agronomists) and end-users (e.g., farmers and agricultural students):

8.2.1. User Feedback:

- Strengths:
 - o Found image detection results visually clear and informative.
 - o Appreciated that detection was fast and results were instant.
 - o Liked the simplicity of the image capture and result viewing process.
- Suggestions:
 - o Requested detection explanation text to understand results better.
 - o Recommended larger icons for better visibility under sunlight.

8.2.2. Technical Feedback :

- Strengths:
 - o High accuracy in bounding box placement.
 - o Efficient model performance even on limited network bandwidth.
 - o Clean UI and functional design appreciated.
- Suggestions:
 - o Include support for offline inference via on-device TensorFlow Lite model.
 - o Enable metadata export of detection results for field analysis.
 - o Add weed infestation severity categorization (e.g., low, medium, high).

8.3 SUMMARY OF TESTING AND FEEDBACK PHASE

The Test and Feedback phase was essential for evolving the weed detection system from a functional prototype to a user-ready solution. Insights gained from both technical and non-technical users were translated into targeted improvements across the backend, model, and mobile interface.

- **Functional Validation:** System accurately identified and highlighted weeds in real field images.
- **Performance Optimization:** Reduced detection time and improved responsiveness.
- **User-Centric Enhancements:** Interface simplified for non-technical users working in outdoor environments.

This stage confirmed the practical viability of the solution and laid the groundwork

for larger-scale pilot testing and deployment in agricultural extension activities.

CHAPTER 9

RE-DESIGN AND IMPLEMENTATION: REFINING THE PROTOTYPE AND PLANNING FOR FUTURE DEVELOPMENT

9.1 INTRODUCTION TO THE REDEFINE AND IMPLEMENTATION PHASE

The Redefine and Implementation phase represents a vital step in transforming the AI-based weed detection prototype into a deployable and user-friendly solution. Building upon earlier stages of design and prototyping, this phase focuses on evaluating performance in real-world agricultural scenarios, refining the system's usability, and planning for feature upgrades. The implementation phase not only ensures that the application meets technical expectations such as accuracy and speed, but also addresses real needs encountered by farmers, agronomists, and field researchers. The improvements made here pave the way for a robust mobile solution that integrates AI with agriculture in an intuitive and accessible manner.

9.2 GATHERING USER FEEDBACK ON THE INITIAL PROTOTYPE

Once the initial version of the weed detection system was functional, feedback was collected from a range of stakeholders including smallholder farmers, agricultural extension officers, and academic researchers. Field testing helped reveal practical challenges and user expectations. Farmers appreciated the ability to detect weeds using only a smartphone camera, but pointed out issues such as slow image processing under poor lighting, and confusion when results were not clearly marked. Users also suggested improvements to the interface, including better visibility of detection results and clearer instructions for capturing images effectively. This feedback was essential for refining the system to match field conditions and user habits.

9.3 ITERATIVE REFINEMENT OF UI AND FUNCTIONALITIES

Based on feedback, several refinements were made to both the UI and core functionality of the application. The user interface was redesigned to be more intuitive, with clearer navigation, enlarged preview areas, and simplified image submission steps. To address detection clarity, bounding boxes were made more prominent and overlaid with weed type and confidence levels. The FastAPI backend was optimized for faster response times and better handling of edge cases, especially in low-contrast image scenarios. Additionally, guidance messages and tooltips were added to help users capture optimal images and understand detection results more easily.

9.4 ENHANCING SYSTEM PERFORMANCE AND STABILITY

Performance optimization was a central focus in this phase. The EfficientDet-D1 model was pruned and fine-tuned for mobile-friendly inference without sacrificing accuracy. Backend latency was minimized through optimized image preprocessing and asynchronous request handling. The system's compatibility across Android devices and screen sizes was tested rigorously, and bugs affecting camera performance or image upload were resolved. Offline capability was validated using a bundled TensorFlow Lite version of the model, enabling weed detection in areas with no internet access. Stability improvements also included caching the last image result and introducing error handling for poor-quality image inputs.

9.5 PLANNING FOR FUTURE FEATURES AND UPGRADES

With a stable and refined prototype in place, the roadmap for future development was outlined. Planned features include multi-class weed recognition to differentiate between various weed species and crop types, integration of GPS tagging to support geospatial weed mapping, and multilingual support to assist regional farmers. A dashboard-style history panel to track past scans and infestation trends is also under consideration. Long-term goals include full offline capability via an on-device model for faster processing, integration with agricultural databases for real-time advisory suggestions, and eventual deployment as a standalone mobile app available on app stores. Collaboration with agricultural institutions is also envisioned to validate and expand the system's usability across diverse farming contexts.

CHAPTER 10

CONCLUSION

The AI-Powered Weed Detection System marks a transformative step in modernizing agricultural practices by leveraging deep learning and mobile technology to assist in real-time weed identification. Designed with a user-centered approach, the system empowers farmers and agricultural professionals with an intuitive, offline-capable solution for early detection and management of weed infestations, ultimately contributing to higher crop yields and sustainable farming.

From the early stages of development, the primary focus remained on building a practical and accessible tool that could function effectively in the diverse and challenging conditions of agricultural fields. By integrating an EfficientDet-D1 model for object detection and deploying a Flutter-based mobile application connected to a FastAPI backend, the system provides fast, accurate, and visual weed detection using a smartphone camera.

Key takeaways from the project include:

- **User-Centric Design:** The interface was designed specifically for farmers and field users, prioritizing simplicity, readability in outdoor lighting, and minimal training requirements. Key functionalities, such as real-time image scanning and result visualization, were crafted to support rapid and actionable decision-making.
- **Technology Integration:** The system harnesses the power of EfficientDet-D1 for high-accuracy weed detection and leverages Flutter and FastAPI for a lightweight, scalable frontend-backend architecture. The detection pipeline operates within seconds, and planned offline capability ensures usability in remote farming locations with limited connectivity.

- **Iterative Refinement:** Using the Stanford Design Thinking framework, the solution evolved through repeated prototyping, feedback loops, and testing under real agricultural conditions. Each iteration helped improve model accuracy, UI clarity, and end-user satisfaction, ensuring a product that meets both technical and practical expectations.
- **Practicality and Usability:** The mobile app facilitates easy image capture and instant results, making it a highly effective tool for routine farm inspections. The backend processes are optimized for speed and efficiency, ensuring minimal lag and maximum reliability in live usage. The annotated image output, featuring bounding boxes around weeds, aids farmers in making timely decisions about weed control.

In conclusion, this project showcases how artificial intelligence, when thoughtfully applied, can address long-standing challenges in agriculture. By combining real-time computer vision with a mobile-first design, the AI-Powered Weed Detection System stands as a scalable, accessible, and impactful solution—poised to assist farmers in reducing manual labor, minimizing chemical usage, and promoting precision farming practices.

CHAPTER 11

FUTURE WORK

While the AI-Powered Weed Detection System successfully addresses the core challenges of real-time weed identification in agricultural environments, there remains significant potential to enhance its functionality, scalability, and user experience. The following outlines the key directions for future development:

1. Offline Functionality Optimization:

- **Full Offline Detection Pipeline:** Improve the system's ability to function entirely offline by embedding the detection model on-device, eliminating the need for server inference and enabling seamless usage in remote farming locations.
- **Model Compression:** Apply techniques like quantization and pruning to reduce the model size and enhance inference speed on low-end mobile devices.

2. Crop-Weed Classification:

- **Multi-Class Detection:** Extend the system to distinguish between different crop types and various weed species, providing richer data for farmers to make targeted weed control decisions.
- **Crop Health Assessment:** Integrate AI capabilities to detect signs of disease or stress in crops alongside weed detection.

3. Integration with IoT and Smart Farming Tools:

- **Drone and Robot Integration:** Allow the system to work with drones or autonomous field robots for large-scale weed detection and automated spraying.
- **Sensor Data Fusion:** Combine weed detection results with data from soil sensors, weather stations, and irrigation systems to support holistic farm management.

4. Advanced Detection Capabilities:

- **Stage-Wise Weed Growth Identification:** Train the model to identify weeds at different stages of growth to support more precise and timely intervention.
- **Density Mapping:** Generate weed infestation heatmaps to help farmers understand spatial distribution and prioritize treatment areas.

5. Farmer Analytics and Dashboards:

- **Infestation History Tracking:** Develop a user dashboard that logs previous scans, weed types detected, and action taken, enabling farmers to track weed trends over time.
- **Farm-Wide Reports:** Allow users to compile data from multiple fields into detailed seasonal weed reports, aiding long-term planning and resource allocation.

6. Expanded Dataset and Model Retraining:

- **Diverse Geographical Dataset:** Enhance model accuracy by training on weed images from diverse soil types, climates, and crop ecosystems across different regions.
- **Active Learning Loop:** Incorporate user-labeled images into the training pipeline to continuously refine model performance over time.

7. Community and Expert Feedback Integration:

- **Agronomist Feedback Feature:** Enable agricultural experts to annotate results or suggest corrections, improving the system's detection reliability and educating farmers.
- **User Community Portal:** Create a shared platform where users can upload field results, discuss detection anomalies, and share weed management strategies.

8. Cross-Platform Support and Multi-Language Interface:

- **iOS and Web App Versions:** Develop support for other platforms including iOS and web browsers to expand accessibility.
- **Multi-Language UI:** Add support for regional languages to cater to farmers from different linguistic backgrounds.

9. In-App Weed Management Guidance:

- **Herbicide Recommendation System:** Based on the type and density of weeds detected, suggest suitable herbicides or manual removal strategies tailored to the crop type.
- **Cost Estimation Tool:** Allow farmers to estimate the cost of treatment and potential yield loss from infestation levels.

10. AI-Powered Notifications and Scheduling:

- **Smart Alerts:** Notify users of optimal times for scanning based on weather conditions and weed growth cycles.
- **Crop Calendar Integration:** Integrate a crop calendar to guide farmers on the best scanning and weeding periods throughout the crop lifecycle.

11. Government and Research Collaboration:

- **Policy Integration:** Collaborate with agricultural departments to create systems that can report weed infestation statistics regionally and influence policy.
- **Academic Research Support:** Provide weed detection data to agricultural researchers to study weed behavior, evolution, and control effectiveness over time.

12. Accessibility Features:

- **Voice Interface:** Integrate speech-based commands for farmers with literacy barriers or physical impairments.
- **Simplified Visuals:** Introduce high-contrast modes and larger UI elements to support users with visual impairments.

By implementing these advancements, the AI-Powered Weed Detection System can evolve into a comprehensive decision-support tool for smart farming. Its scalability, offline readiness, and real-world applicability position it as a valuable innovation not only for individual farmers but also for agri-tech companies, government agricultural departments, and rural development initiatives—helping to drive the future of sustainable agriculture practices.

CHAPTER 12

REFLECTING ON THE VALUE OF DESIGN THINKING IN THIS RESEARCH

12.1 EMBRACING A HUMAN-CENTERED MINDSET

One of the key realizations during this project was the importance of applying a human-centered mindset in developing technological solutions for agriculture. Design Thinking offered a structured yet empathetic approach to understanding the real needs of farmers and agricultural workers. Rather than focusing solely on model performance or technical implementation, this mindset allowed the team to explore how the weed detection system could genuinely support the daily struggles faced in the field. It shifted the goal from simply detecting weeds to building a practical, accessible, and intuitive tool that enhances farming efficiency and decision-making.

12.2 LEARNING THROUGH ITERATION AND FEEDBACK

Throughout the development of the prototype, the iterative nature of Design Thinking proved to be essential. Early versions of the system were shared with real users—farmers, agronomists, and agricultural students—whose feedback guided refinements in both the model’s detection capability and the user interface. From image clarity to detection labeling, every round of feedback contributed to a more polished and user-friendly product. This cycle of building, testing, and learning reinforced the importance of adaptability and continuous improvement in

real-world problem solving..

12.3 FOSTERING CROSS-DISCIPLINARY THINKING

This project brought together multiple fields: artificial intelligence, mobile development, user interface design, and agricultural science. Design Thinking facilitated a cross-disciplinary mindset, encouraging collaboration between technical developers and domain experts in farming. This holistic view allowed for better decision-making—such as balancing model complexity with mobile device performance or designing the UI with consideration for varying literacy levels. It helped ensure the technology was not just accurate, but also usable in rural, offline environments..

12.4 ENHANCING PROBLEM FRAMING AND CLARITY

The Define phase of Design Thinking played a pivotal role in clarifying the true problem: not merely detecting weeds, but enabling early intervention, improving crop health, and reducing manual labor for farmers. This clear framing led to a more targeted and meaningful solution. Instead of building a generic AI tool, the project was shaped around specific needs such as offline usage, image-based detection with bounding boxes, and easy result interpretation. This focus guided the technical and design decisions throughout the implementation process..

12.5 ENCOURAGING CREATIVITY AND INNOVATION

The Ideate phase encouraged the exploration of bold, forward-thinking features without immediately worrying about feasibility. This mindset led to the inclusion of unique elements such as weed density estimation, real-time field scanning using mobile cameras, and

integration of localized weed data. By creating a space for creativity early on, the project expanded beyond its initial scope and evolved into a smart agricultural assistant with real potential for scalability. This creativity, nurtured through Design Thinking, made the solution not only functional but also innovative and impactful.

CHAPTER 13

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